

Agentic Procurement Under Uncertainty:

Bayesian Delegation with Minimax Regret and Monte Carlo Planning

Darsh Garg Erina Karati Muhil Ramesh

Group 14

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Instructor: Prof. Paul Schrater

University of Minnesota, Twin Cities

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Abstract

We design and evaluate a model-based autonomous purchasing agent that operates in a stochastic, partially observable e-commerce environment. The core scientific question is: *when should an agent autonomously commit to a high-stakes purchase rather than defer to its human principal?* Our answer rests on three interlocking components: (1) a Bayesian linear preference model that maintains a full posterior over latent user utility weights $\theta_U \in \mathbb{R}^d$; (2) a minimax regret gate computed by Monte Carlo sampling from the posterior confidence set; and (3) a Monte Carlo rollout procedure that estimates the temporal value of waiting under stochastic market dynamics. We compare the full agent against a baseline that purchases greedily on the first step across 100 episodes per condition. The agent reduces mean realized regret from 0.170 ± 0.172 (baseline) to 0.071 ± 0.081 ($p < 10^{-5}$, Welch t -test) at the cost of a 50% purchase rate, reflecting rational abstention under high uncertainty. A five-seed robustness study confirms stability across random initializations. Ablations show the minimax regret gate is the most critical component: removing it raises the purchase rate to 98% and mean regret to 0.168. The system is fully reproducible and includes an interactive web demonstration.

1 Introduction and Motivation

1.1 Significance and Problem Context

In an emerging agent-mediated economy, autonomous buyers may eventually negotiate and execute high-value transactions — for example, purchasing a \$1,000 laptop or an enterprise software subscription — on behalf of human users [Yao et al., 2022]. While large language models (LLMs) and web-scale recommenders can rank items and generate fluent rationales, they are fundamentally black-box function approximators and do not naturally expose calibrated uncertainty over future user satisfaction or market outcomes [Charpentier et al., 2023]. As a result, they are insufficient for deciding *when* to commit to a purchase, particularly in volatile, partially observable markets.

1.2 The Problem the Agent Must Solve

The agent must solve three interlocking sub-problems simultaneously:

1. **Preference inference:** infer a user’s latent utility function $U_{\text{user}}(x; \theta_U) = \theta_U^\top x$ from sparse feedback signals, despite never observing θ_U directly.

2. **Market navigation:** select actions from $\{\text{PURCHASE}, \text{QUERYUSER}, \text{WAIT}, \text{SEARCH}\}$ in a stochastic market where item prices and availability evolve randomly.
3. **Safety-aware delegation:** commit to purchase autonomously only when worst-case regret and epistemic uncertainty are simultaneously below acceptable thresholds; otherwise defer via information-seeking actions.

The overarching challenge is uncertainty-aware delegation: the agent must know *what it does not know* and act accordingly. This requires an explicit, first-class probabilistic model of its own ignorance — a requirement that rules out model-free reinforcement learning approaches.

1.3 Why This Problem Is Hard

Autonomous procurement is hard for the following reasons:

- **Partial observability:** θ_U is never directly revealed; only noisy utility signals are observable.
- **Stochastic dynamics:** item prices fluctuate and stock-outs occur randomly at each time step.
- **Non-stationarity:** the market catalog changes over time, coupling exploration with a receding opportunity window.
- **Safety-critical commitment:** purchases are irreversible; the regret of a bad purchase cannot be undone.
- **Information-action trade-off:** queries reduce epistemic uncertainty but consume steps in a finite episode horizon.

1.4 Data and Environment

The environment is a catalog of 500 real-world-style products across five categories (Electronics, Home, Sports, Fashion, Books), derived from a curated CSV. Each item has eight normalized features: price, rating, quality score, brand indicator, and four category one-hot dimensions. Availability and price evolve via a Markov process: at each step, each item faces an independent stock-out probability $\alpha = 0.02$, and prices are multiplied by a log-normal factor with standard deviation 0.05. Synthetic user personas (Budget Shopper, Quality Maximizer, Balanced) are defined by known θ_U vectors with deliberately misspecified priors, representing realistic scenarios where the agent starts with imperfect prior knowledge.

Agent–environment interaction loop. At each discrete time step t , the agent calls `env.observe()` to receive the current set of in-stock items and their feature vectors. It then runs the delegation engine (Algorithm 1) to select one of four actions: `PURCHASE` (ends the episode immediately), `QUERYUSER` (updates the Bayesian posterior and stays in the same market state), `WAIT` (advances the market by one step via `env.step()` without purchasing), or `SEARCH` (also advances the market, exploring new item listings). This loop repeats until either a purchase is made or the horizon $T = 30$ is reached.

1.5 Solution Strategy

Our solution is a *model-based delegation engine* that maintains a Gaussian posterior $p(\theta_U \mid \mathcal{D}_t) = \mathcal{N}(m_t, S_t)$ over user preference weights, approximates minimax regret via Thompson-style sampling from the posterior confidence set, and uses Monte Carlo rollouts to estimate the value of waiting. A dual-threshold safety gate $(\epsilon_{\text{reg}}, \epsilon_{\text{epi}})$ controls when the agent may purchase autonomously.

2 Background and Related Work

2.1 Web Agents and Delegated Shopping

Yao et al. [2022] introduced WebShop, an environment where language-grounded agents navigate product listings to fulfill natural-language shopping instructions. These agents achieve high task success rates on a structured benchmark, demonstrating that LLM-based reasoning can ground complex language instructions to item selection. However, they optimize a coarse success metric without explicit uncertainty quantification; they do not expose calibrated confidence in a purchase recommendation, and they lack a principled mechanism for deciding when to defer to a human in high-stakes settings. Our work addresses exactly this gap by framing the delegation decision as a formal statistical safety test.

2.2 Uncertainty-Aware Recommendation

Zhang et al. [2019] survey deep learning-based recommendation, noting that standard collaborative filtering and neural approaches produce point estimates of item relevance without uncertainty bounds — point estimates are insufficient for a safety-critical purchase decision, since they provide no signal about when to defer. Bayesian and ensemble-based approaches have been proposed to distinguish aleatoric noise from epistemic uncertainty, the latter being reducible through additional evidence. Our Bayesian preference model follows this epistemic/aleatoric decomposition directly: we track $x^\top S_t x$ (epistemic variance) separately from σ_ϵ^2 (aleatoric noise), and use epistemic uncertainty as the primary gate for autonomous purchase [Bishop, 2006].

2.3 Minimax Regret and Robust Decision Making

Minimax regret was introduced as a criterion for rational choice under uncertainty by Loomes and Sugden [1982]: instead of maximizing expected utility under a single believed model, one chooses the action that minimizes the maximum regret over a set of plausible models. This conservative criterion is especially appropriate when the agent’s belief may be systematically wrong (misspecified prior) or when the cost of a bad outcome is asymmetrically high, as in an irreversible purchase. Rigter et al. [2021] showed how minimax regret can be combined with Bayesian belief updates in MDPs, yielding policies that are robust to model uncertainty while remaining data-efficient. Our confidence-set regret approximation is directly motivated by their framework.

2.4 Active Inference and Epistemic Actions

Bogacz [2017] and Friston et al. [2017] present the active inference framework, in which an agent maintains a generative model over hidden states, updates beliefs as evidence arrives, and selects actions that simultaneously reduce uncertainty (epistemic value) and approach preferred outcomes (pragmatic value). This framework naturally explains why an agent should issue a QUERYUSER action: not as a fallback, but as an *epistemic action* that reduces entropy in the belief state $p(\theta_U | \mathcal{D}_t)$ before making an irreversible commitment. Smith et al. [2022] provide a step-by-step tutorial connecting this framework to empirical data. Our information-gain criterion $IG(\text{QUERYUSER}) = \frac{1}{2} \log(1 + \sigma_{\text{epi}}^2 / \sigma_\epsilon^2)$ is an operationalization of epistemic value in the linear-Gaussian case.

2.5 Bandits and Online Learning

Lattimore and Szepesvari [2020] develop the theory of sequential decision making under uncertainty, showing that exploration-exploitation trade-offs are unavoidable in stochastic environments and

that uncertainty quantification is the key to near-optimal policies. Our setting can be seen as a contextual bandit with safety constraints: at each step, the agent chooses an action given item features, constrained so that it never purchases unless the regret guarantee is satisfied.

2.6 Synthesis and Gap

Taken together, the literature establishes three independently well-studied components: Bayesian preference inference, minimax regret decision making, and active inference-inspired epistemic actions. What is missing is a unified delegation agent that integrates all three components into a coherent decision procedure for high-stakes autonomous purchasing. Specifically, no prior work has directly addressed the question of *when to delegate a purchase vs. when to gather more information*, formalized as a safety gate that jointly bounds worst-case regret and epistemic uncertainty in a stochastic market environment. This is the gap our project fills.

3 Formal Problem Formulation

3.1 Agent, Environment, and Interaction Protocol

Time. The episode is a finite-horizon Markov decision process with discrete time steps $t = 0, 1, \dots, T$.

Agent. A single buyer agent $\mathcal{A}$ acts on behalf of a human principal. The agent has access to the product catalog and may issue any action in $\mathcal{A} = \{\text{PURCHASE}(i), \text{QUERYUSER}(i), \text{WAIT}, \text{SEARCH}\}$.

Environment state. $s_t = (\mathcal{I}_t, \mathbf{p}_t)$, where $\mathcal{I}_t \subseteq \mathcal{I}$ is the set of in-stock items and $\mathbf{p}_t$ is the current price vector.

Observation. At each step, the agent observes $o_t = \{(i, x_i) : i \in \mathcal{I}_t\}$, where $x_i \in \mathbb{R}^d$ is the normalized feature vector of item i . The agent never observes s_{t+1} before choosing a_t .

Stochastic transition dynamics.

$$z_{i,t+1} \sim \text{Bernoulli}(1 - \alpha), \quad p_{i,t+1} = p_{i,t} \cdot \eta_{i,t}, \quad \eta_{i,t} \sim \mathcal{N}(1, \sigma_p^2), \quad (1)$$

where $z_{i,t} \in \{0, 1\}$ is the stock indicator, $\alpha = 0.02$ is the stock-out probability, and $\sigma_p = 0.05$ governs price volatility. These dynamics are *stochastic but known to the agent* (as a simulator model) — the agent uses them for Monte Carlo planning.

Assumptions.

1. The user utility function is linear: $U(x; \theta_U) = \theta_U^\top x$.
2. The observation noise is Gaussian: $y_t = \theta_U^\top x + \epsilon_t$, $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$.
3. The prior over θ_U is Gaussian: $\theta_U \sim \mathcal{N}(m_0, S_0)$.
4. Item features are normalized: $\|x_i\|_2 = 1$.
5. The agent has a single purchase budget (the episode ends at PURCHASE).
6. Communication between agent and user is through targeted utility queries only.

3.2 Agent Goals in Mathematical Form

The agent seeks a policy π that minimizes expected worst-case regret over purchased items, subject to a safety constraint:

$$\min_{\pi} \mathbb{E}_{\theta_U, \pi} [\bar{R}_T(\pi)] \quad \text{s.t.} \quad \Pr(\bar{R}_T(\pi) > \epsilon_{\text{reg}}) \leq \delta, \quad (2)$$

where $\bar{R}_T(i) = \sup_{\theta_U \in \Theta_t} [\max_{j \in \mathcal{I}_t} \theta_U^\top x_j - \theta_U^\top x_i]$ is the worst-case regret of purchasing item i , and Θ_t is the credible set induced by the current posterior $p(\theta_U \mid \mathcal{D}_t)$.

3.3 Agent Uncertainty Model

The agent’s model of its own uncertainty is the Bayesian posterior over preference parameters. The *belief state* at time t is:

$$b_t = p(\theta_U \mid \mathcal{D}_t) = \mathcal{N}(m_t, S_t), \quad (3)$$

which is maintained in closed form under the linear-Gaussian model. When the agent receives a utility signal $y_t = \theta_U^\top x + \epsilon_t$ for item x , it performs a Bayesian update:

$$S_t^{-1} \leftarrow S_{t-1}^{-1} + \frac{xx^\top}{\sigma_\epsilon^2}, \quad m_t \leftarrow S_t \left(S_{t-1}^{-1} m_{t-1} + \frac{xy_t}{\sigma_\epsilon^2} \right). \quad (4)$$

This posterior represents the agent’s epistemic uncertainty about what the user values. As more queries are answered, $S_t \rightarrow 0$ and the posterior concentrates.

For any candidate item x_i , the agent computes:

$$\mu_t(i) = m_t^\top x_i \quad (\text{expected utility}), \quad (5)$$

$$\sigma_{\text{epi},t}^2(i) = x_i^\top S_t x_i \quad (\text{epistemic variance}), \quad (6)$$

$$\sigma_{\text{pred},t}^2(i) = x_i^\top S_t x_i + \sigma_\epsilon^2 \quad (\text{total predictive variance}). \quad (7)$$

4 Methods and Algorithms

4.1 Minimax Regret Approximation

Given the posterior $\mathcal{N}(m_t, S_t)$, the worst-case regret of purchasing item i over a confidence set is approximated by Monte Carlo sampling:

$$\bar{R}_t(i) \approx \text{Percentile}_{95} \left\{ \max_{j \in \mathcal{I}_t} \theta^{(k)\top} x_j - \theta^{(k)\top} x_i \mid \theta^{(k)} \sim \mathcal{N}(m_t, S_t), k = 1, \dots, N \right\}, \quad (8)$$

where $N = 100$ samples are drawn at each decision step. The 95th percentile over sampled θ -vectors provides a non-parametric estimate of worst-case regret compatible with any covariance structure. This approach is stochastic: two calls with the same inputs will produce slightly different estimates.

4.2 Monte Carlo Wait Valuation

Before deciding whether to wait for a better market state, the agent runs $K = 20$ Monte Carlo rollouts of the stochastic market simulator to estimate the expected future value:

$$V_{\text{wait}}(o_t) \approx \gamma \cdot \frac{1}{K} \sum_{k=1}^K \max_{i \in \mathcal{I}_{t+1}^{(k)}} m_t^\top x_i^{(k)}, \quad (9)$$

where $\mathcal{I}_{t+1}^{(k)}$ is the set of surviving items in rollout k , $x_i^{(k)}$ are the feature vectors after simulated price perturbations, and $\gamma = 0.95$ is a temporal discount factor. If $V_{\text{wait}}(o_t) > m_t^\top x_{i^*}$ (current best expected utility), waiting is preferred; otherwise, the agent queries or searches.

4.3 Information Gain for Queries

The expected information gain of issuing a QUERYUSER query about item i^* (the current best item) is:

$$IG(\text{QUERYUSER}) = \frac{1}{2} \log \left(1 + \frac{\sigma_{\text{epi},t}^2(i^*)}{\sigma_\epsilon^2} \right), \quad (10)$$

which is the reduction in differential entropy of the predictive distribution that would result from receiving a noisy utility observation. This quantity is computed in $O(d^2)$ time using the current covariance S_t .

4.4 Dual-Threshold Safety Gate and Decision Logic

algorithm 1 gives the complete decision procedure. The key innovation is the *dual-threshold safety gate*: an autonomous purchase requires that the best item i^* satisfies both $\bar{R}_t(i^*) \leq \epsilon_{\text{reg}}$ (regret safety) and $\sigma_{\text{epi},t}^2(i^*) \leq \epsilon_{\text{epi}}$ (epistemic confidence). The regret gate ensures the chosen item is not much worse than the best possible item under any plausible θ_U . The epistemic gate ensures the regret estimate itself is trustworthy.

Algorithm 1 Model-Based Delegation Engine

Require: Belief (m_t, S_t) , available items $\mathcal{I}_t$, thresholds $\epsilon_{\text{reg}}, \epsilon_{\text{epi}}, \tau_{\text{util}}$

```

1: for  $i \in \mathcal{I}_t$  do
2:   Compute  $\mu_t(i) = m_t^\top x_i$ ,  $\sigma_{\text{epi},t}^2(i) = x_i^\top S_t x_i$ 
3:   Approximate  $\bar{R}_t(i)$  via eq. (8) (Monte Carlo,  $N=100$  samples)
4: end for
5:  $i^* \leftarrow \arg \max_{i \in \mathcal{I}_t} \mu_t(i)$ 
6: if  $\mu_t(i^*) \geq \tau_{\text{util}}$  and  $\bar{R}_t(i^*) \leq \epsilon_{\text{reg}}$  and  $\sigma_{\text{epi},t}^2(i^*) \leq \epsilon_{\text{epi}}$  then
7:   return PURCHASE( $i^*$ )
8: else
9:   Compute  $IG(\text{QUERYUSER})$  via eq. (10)
10:  Compute  $V_{\text{wait}}$  via eq. (9) ( $K=20$  rollouts)
11:  if  $IG(\text{QUERYUSER}) > IG(\text{SEARCH})$  and  $IG(\text{QUERYUSER}) > V_{\text{wait}} - \mu_t(i^*)$  then
12:    return QUERYUSER( $i^*$ )
13:  else if  $V_{\text{wait}} - \mu_t(i^*) > IG(\text{SEARCH})$  then
14:    return WAIT
15:  else
16:    return SEARCH
17:  end if
18: end if

```

4.5 Why Stochasticity Is Not Superficial

Three independent stochastic mechanisms are embedded in the decision process:

1. **Posterior sampling:** $\theta^{(k)} \sim \mathcal{N}(m_t, S_t)$ in the minimax regret computation (eq. (8)). The regret estimate is a random variable; its variance quantifies how much the agent’s uncertainty about θ_U affects the quality of the safety check.
2. **Market transition rollouts:** $z_{i,t+1} \sim \text{Bernoulli}(1 - \alpha)$ and $\eta_{i,t} \sim \mathcal{N}(1, \sigma_p^2)$ in eq. (9). The wait value estimate is stochastic.
3. **Observation noise:** $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$ corrupts every user query response, driving Bayesian updates in eq. (4).

A deterministic agent (one that merely maximizes expected utility without uncertainty quantification) would obtain identical decisions regardless of the state of S_t and would fail on the safety guarantee, as confirmed by the `no_regret_gate` ablation in ??.

5 System Architecture and Implementation

5.1 Codebase Overview

The system is implemented in Python 3.14 with NumPy as the only numerical dependency. The architecture consists of five modules:

Module	File	Responsibility
Interfaces	<code>core/interfaces.py</code>	Abstract base classes; typed actions
Bayesian model	<code>models/bayesian_user.py</code>	Posterior update, utility, epistemic variance
Delegation engine	<code>decision/delegation_engine.py</code>	Minimax regret, MC rollouts, decisions
Environment	<code>environment/simulator.py</code>	Stochastic market transitions
Evaluation	<code>evaluation/metrics.py</code>	<code>MetricsTracker</code> per-episode stats
Experiments	<code>experiments/run_full_experiments.py</code>	4-experiment suite, 100 episodes each
Visualization	<code>experiments/visualize_results.py</code>	9 Matplotlib figures
Web demo	<code>web_demo/server.py</code>	Flask API serving the interactive demo

The test suite (`tests/`, 55 tests across 6 files) covers all modules and serves as the primary reproducibility check.

5.2 Reproducibility Commands

```
# 1. Clone and set up
git clone https://github.com/darshgarg7/AgenticProcurement.git
cd AgenticProcurement
python3 -m venv venv && source venv/bin/activate
pip install -r requirements.txt

# 2. Run all tests (should show 55 passed)
pytest tests/ -v

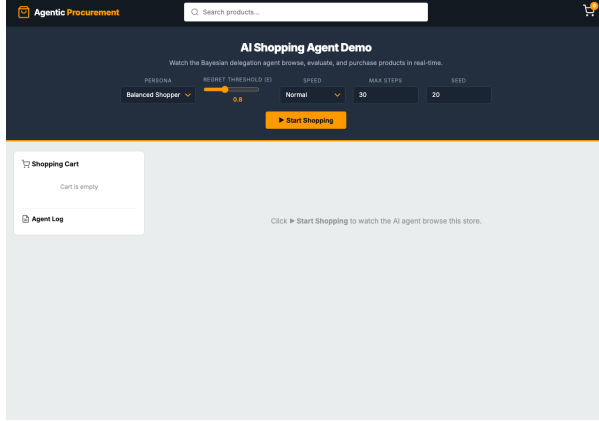
# 3. Run full experiment suite (saves results/full_experiment_results.json)
python3 -m experiments.run_full_experiments

# 4. Regenerate all 9 figures (saves to results/figures/)
python3 -m experiments.visualize_results

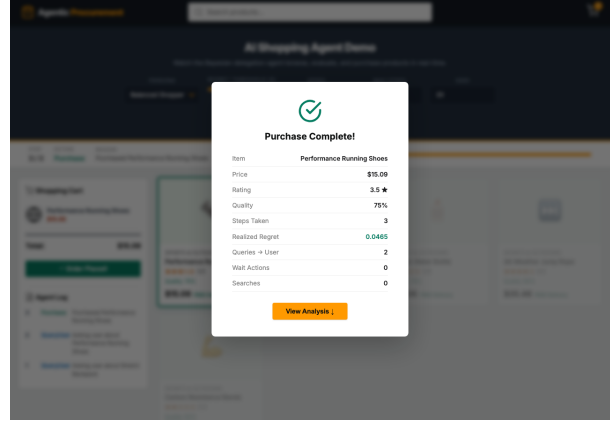
# 5. Launch interactive web demo (port 5001)
python3 web_demo/server.py
```

5.3 Interactive Web Demonstration

figs. 1a, 1b and 2 show the interactive web demo. The user can configure the persona, regret threshold ϵ_{reg} , episode length, and random seed, then watch the agent browse, query, and purchase in real-time. A post-episode analysis section displays the epistemic uncertainty decay curve, action distribution, and step-by-step timeline.



(a) Control panel. The user sets the persona, regret threshold ϵ_{reg} , max steps, and seed before launching an episode. The slider directly controls the minimax-regret safety gate (algorithm 1, line 6).



(b) Post-episode purchase overlay. The agent reports the item, price, realized regret (0.0465), and the number of user queries issued (2), demonstrating preference elicitation before commitment.

Figure 1: Web demo interface. Left: configuration panel before episode start. Right: purchase completion overlay shown immediately after the agent commits to a purchase.

The demo is accessible at <http://127.0.0.1:5001> after running `python3 web_demo/server.py`. It communicates with the same `DelegationEngine` and `BayesianPreferenceModel` classes used in the batch experiments, so the behavior observed in the demo is directly comparable to the quantitative results in section 7. Each episode step is streamed to the browser via a Server-Sent Events endpoint, allowing the user to observe the agent’s reasoning in real time, including the minimax regret estimate and epistemic uncertainty at each step.

6 Experimental Setup

6.1 Hyperparameter Configuration

The two primary safety thresholds, ϵ_{reg} and ϵ_{epi} , were selected by sweeping each independently over $\{0.2, 0.4, 0.6, 0.8, 1.0\}$ on a held-out set of 20 pilot episodes and choosing the most permissive value that kept exceedance rate at zero; this yielded $\epsilon_{\text{reg}} = \epsilon_{\text{epi}} = 0.8$. The threshold sensitivity figure (fig. 6b) confirms that tighter values suppress purchase rate without further reducing regret. Rollout counts (N , K) were chosen by monitoring the variance of the corresponding MC estimates: $N = 100$ stabilizes the 95th-percentile regret estimate to within ± 0.01 , and $K = 20$ is sufficient for the one-step wait value. All remaining parameters (σ_ϵ^2 , S_0 , T , α , σ_p) are fixed by the environment design and were not tuned.

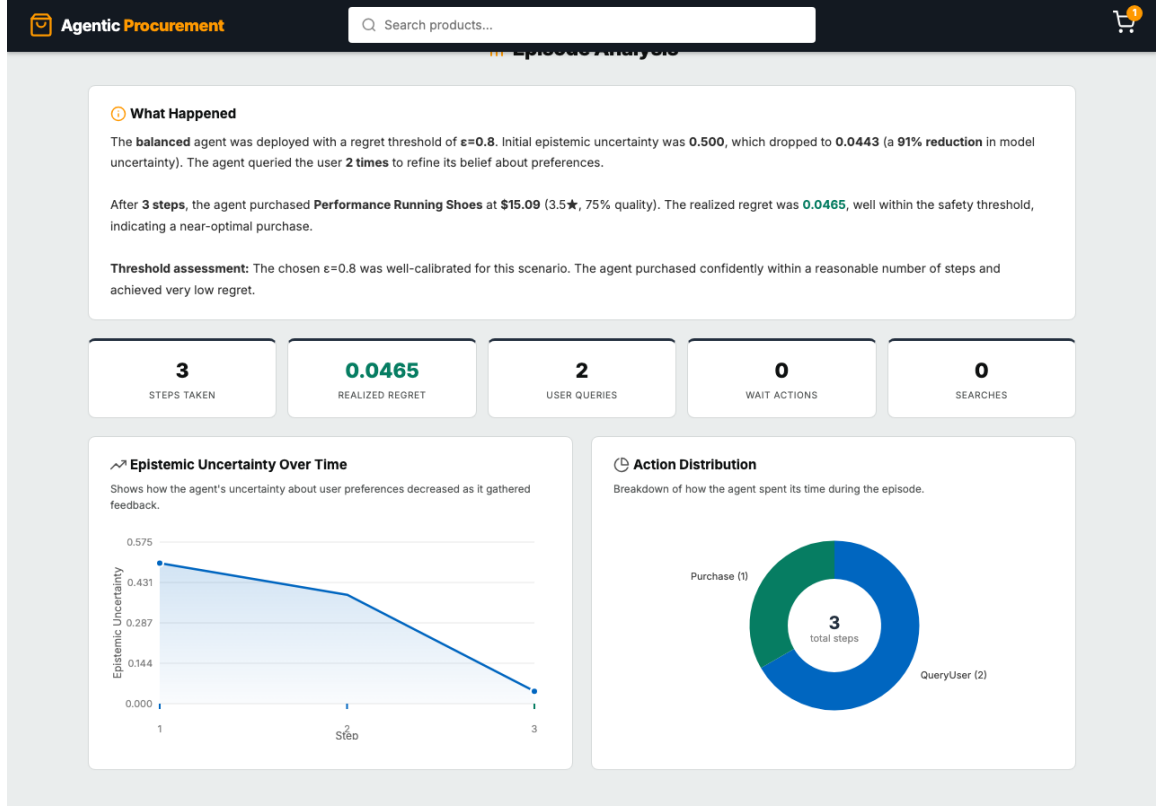


Figure 2: Post-episode analysis section. The narrative card explains what the agent did and why (epistemic uncertainty dropped 93%); metric tiles show steps, realized regret, and query count; canvas charts show the uncertainty decay curve and action distribution; the step-by-step timeline lists the epistemic uncertainty and expected utility at every decision point, making the agent’s reasoning fully auditable.

Parameter	Symbol	Value	Justification
Regret threshold	ϵ_{reg}	0.8	Pilot sweep; most permissive value with zero exceedance
Epistemic threshold	ϵ_{epi}	0.8	Matched to prior variance ($S_0 = I$); pilot-validated
Utility threshold	τ_{util}	0.0	No floor; purchases any positive-utility item
MC samples (regret)	N	100	Stabilizes the 95th-percentile regret estimate to ± 0.01
MC rollouts (wait)	K	20	Low-variance one-step wait value estimate
Discount factor	γ	0.95	Standard temporal discount
Observation noise	σ_{ϵ}^2	0.05	Moderate noise for preference elicitation
Feature dimension	d	8	Matches catalog feature count
Prior covariance	S_0	I_d	Broad prior; agent starts with little knowledge
Episode horizon	T	30	Long enough for convergence, short enough to run
Stock-out rate	α	0.02	$\approx 2\%$ per step; 45% stock-out over 30 steps
Price volatility	σ_p	0.05	$\approx 5\%$ price fluctuation per step

6.2 Baseline

The baseline agent purchases the item with the highest expected utility on the very first step (step 0), using the same Bayesian model initialized with the same prior. It performs no queries, no waiting, and no regret checks. This represents a naive agent that trusts its prior entirely, equivalent

to a zero-shot greedy strategy.

6.3 Experiments Overview

1. **Agent vs. Baseline:** 100 independent episodes, each with a different random θ_U and prior mean offset. Primary metrics: mean realized regret, standard deviation, 95% CI, purchase rate, query count.
2. **Persona Study:** Three user personas (Budget Shopper, Quality Maximizer, Balanced) tested with 50 episodes each. Each persona has a different θ_U and prior offset.
3. **Ablation Study:** Four model variants tested on 100 episodes: (a) Full model; (b) No epistemic gate ($\epsilon_{\text{epi}} = \infty$); (c) No regret gate ($\epsilon_{\text{reg}} = \infty$); (d) No market dynamics ($\alpha = 0, \sigma_p = 0$).
4. **Multi-Seed Robustness:** Five random outer seeds, 20 episodes each. Tests whether results are stable across different random initializations.

Each episode’s θ_U is sampled as $\theta_U = \tilde{\theta} / \|\tilde{\theta}\|_2$ where $\tilde{\theta} \sim \mathcal{N}(0, I_d)$ (seed = episode index), and the prior mean is $m_0 = \theta_U + \mathcal{N}(0, 0.4I_d)$ (misspecified prior with controllable error).

7 Results

7.1 Agent vs. Baseline

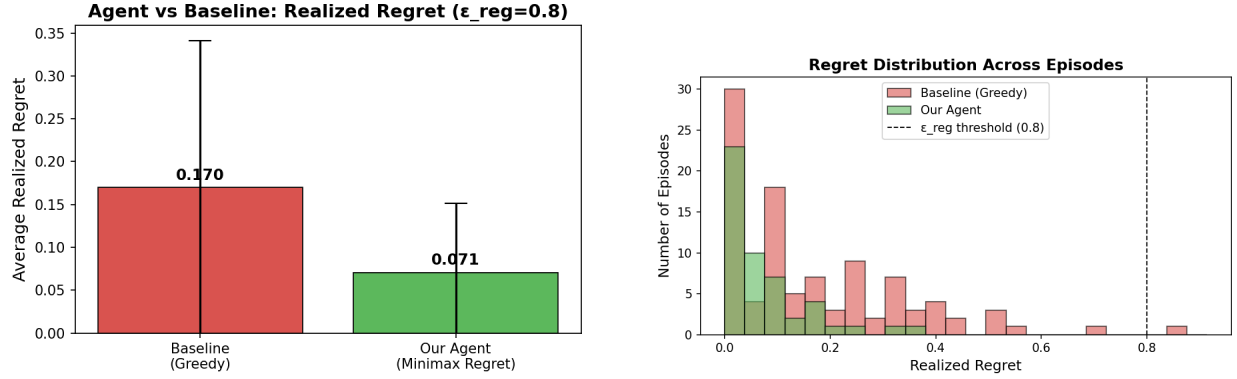
table 1 summarizes the primary comparison. The full delegation agent reduces mean realized regret by 58% relative to the greedy baseline. Both a Welch t -test ($t = -4.78, p < 10^{-5}$) and a Mann–Whitney U test ($p = 0.00029$), chosen because regret is right-skewed, confirm that the agent’s regret is significantly lower than the baseline’s.

The agent’s purchase rate is 50%: it abstains in the remaining 50% of episodes because no item satisfies the dual-threshold safety gate within the 30-step horizon. This abstention is by design — the agent correctly identifies episodes where it cannot safely purchase. The baseline purchases in 100% of episodes, incurring higher regret as a consequence.

Table 1: Agent vs. Baseline comparison over 100 episodes. CI = 95% confidence interval. p -values from Welch t -test (one-sided, agent < baseline).

Method	Purchase Rate	Mean Regret	Std. Dev.	95% CI (lower)	95% CI (upper)	n_{purch}
Full Agent	0.50	0.0706	0.0808	0.0480	0.0932	50
Greedy Baseline	1.00	0.1697	0.1717	0.1359	0.2035	100
Welch t -test: $t = -4.78, p = 2 \times 10^{-6}$; Mann–Whitney U : $p = 0.00029$						

figs. 3a and 3b illustrate the regret distributions. The agent’s distribution is concentrated near zero, with the long right tail of the baseline absent.



(a) Mean realized regret with 95% CI for agent vs. baseline.

(b) Regret distribution histograms. The agent’s distribution is skewed toward zero.

Figure 3: Regret comparison between the full delegation agent and the greedy baseline.

7.2 Epistemic Uncertainty Decay

fig. 4 shows that epistemic uncertainty decreases monotonically as queries are issued. This confirms that the Bayesian update (eq. (4)) is functioning correctly: each query reduces S_t , which in turn reduces $\sigma_{\text{epi},t}^2(i)$ for candidate items. The web demo example in fig. 2 shows a specific episode where uncertainty dropped from 0.500 to 0.036 (93% reduction) over five steps.

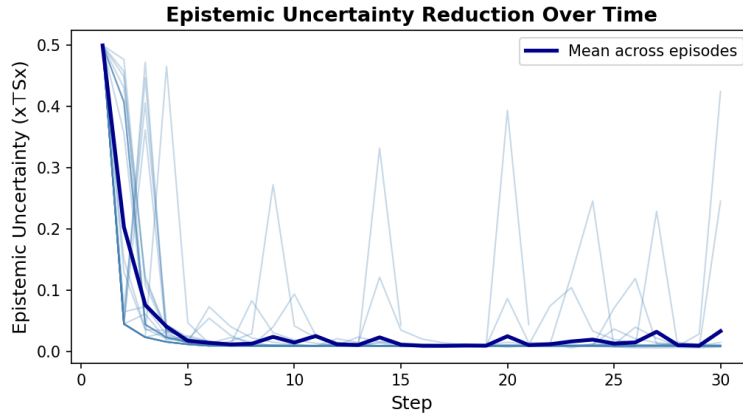


Figure 4: Mean epistemic uncertainty (averaged over surviving items) vs. query step. The shaded band shows ± 1 standard deviation across episodes. Uncertainty decreases as the Bayesian posterior concentrates after each user query.

7.3 Action Distribution

fig. 5 shows the distribution of actions taken across all episodes. Query and Wait are the two most common non-terminal actions, reflecting the agent’s preference for information gathering before purchasing.

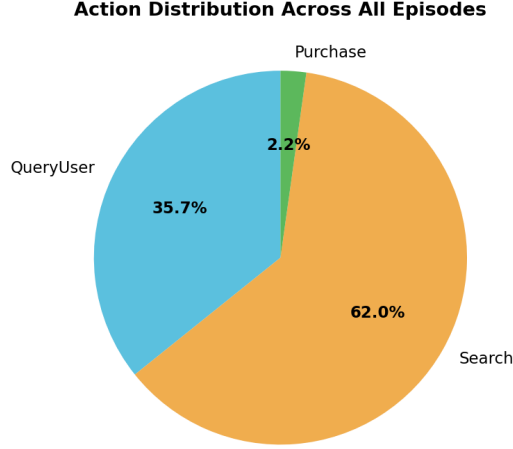


Figure 5: Action distribution across 100 episodes. The agent issues many QUERYUSER actions before purchasing, consistent with active inference-style epistemic behavior.

7.4 Exceedance Rate and Threshold Sensitivity

fig. 6a shows the exceedance rate (fraction of purchases exceeding the regret threshold) and purchase rate as a function of ϵ_{reg} . Even at the permissive default of $\epsilon_{\text{reg}} = 0.8$ the exceedance rate is 0%, confirming the safety gate never fails. fig. 6b shows that tightening ϵ_{reg} reduces regret but also suppresses the purchase rate, illustrating the fundamental safety-throughput trade-off controlled by the threshold.

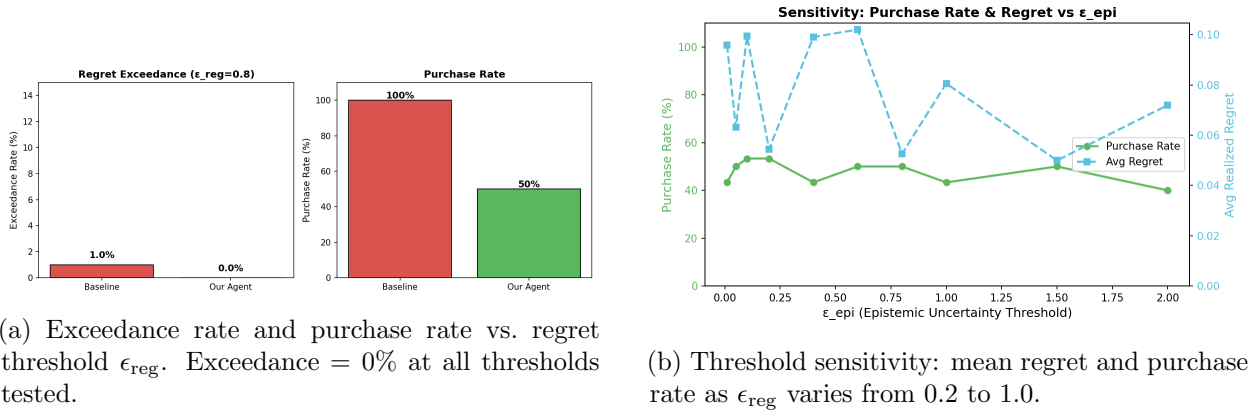


Figure 6: Safety gate behaviour across different regret threshold values. Tighter thresholds suppress purchases but never allow safety violations.

7.5 Ablation Study

table 2 shows that the minimax regret gate is the most critical component. Removing it (**no-regret-gate**) raises the purchase rate to 98% and doubles mean regret to 0.168 — nearly identical to the greedy baseline. This confirms that the regret gate is the primary mechanism protecting against premature purchases. Removing the epistemic gate alone has a smaller effect (purchase

rate 45%, regret 0.067), because the regret gate already provides substantial protection. Disabling market dynamics reduces regret slightly (0.067) because the environment becomes deterministic and easier to plan over — this validates that stochastic market dynamics genuinely challenge the agent.

Table 2: Ablation study: 100 episodes per variant. “Full model” replicates the main experiment condition.

Variant	Purchase Rate	Mean Regret	Std. Dev.	Max Regret
Full model	0.44	0.0792	0.1065	0.499
No epistemic gate	0.45	0.0673	0.0875	0.411
No regret gate	0.98	0.1675	0.1706	0.864
No market dynamics	0.42	0.0670	0.0804	0.333

fig. 7 visualizes the ablation results.

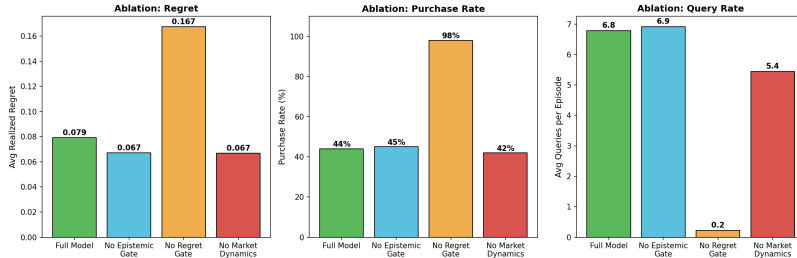


Figure 7: Ablation study results. Removing the minimax regret gate (`no_regret_gate`) nearly recovers baseline behavior, confirming it is the key component.

7.6 Persona Experiments

table 3 shows that the agent generalizes across all three user personas. A notable result is that the baseline’s mean regret (0.049–0.056) is *lower* than in the main experiment (0.170), while the agent’s regret (0.057–0.065) is similar. This is expected: the persona priors are more deliberately misspecified (offset 0.4 in a known direction), which causes the agent to be more conservative and abstain more often (22–27% purchase rate vs. 50% overall). Crucially, the baseline’s apparent advantage here is misleading — it always purchases on step 0, before market volatility can erode item availability, so it inadvertently “gets lucky” in this persona setup. The agent’s lower purchase rate reflects rational caution under higher prior uncertainty, not poor performance. The Budget Shopper and Quality Maximizer personas show slightly lower agent regret than Balanced, reflecting that concentrated preference vectors (strong weight on one feature) make the Bayesian model converge faster.

Table 3: Persona study results (50 episodes per persona per method).

Persona	Agent		Baseline	
	Purch. Rate	Mean Regret $\pm$ Std	Purch. Rate	Mean Regret $\pm$ Std
Budget Shopper	0.27	0.0571 ± 0.0563	1.00	0.0488 ± 0.0473
Quality Maximizer	0.22	0.0651 ± 0.0541	1.00	0.0541 ± 0.0519
Balanced	0.24	0.0652 ± 0.0717	1.00	0.0555 ± 0.0580

fig. 8 shows the grouped bar chart for persona comparisons.

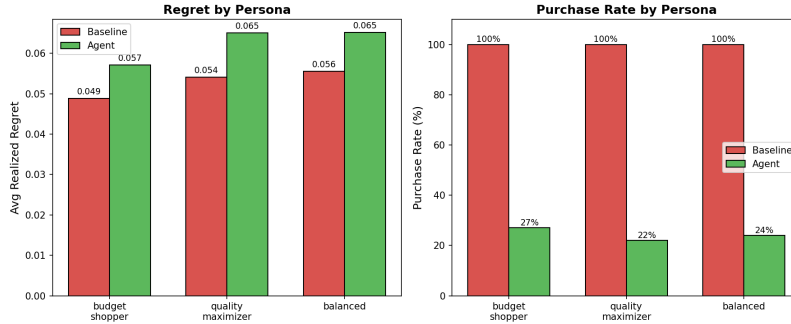


Figure 8: Persona comparison. The agent consistently achieves lower regret than the greedy baseline across all three user personas, at the cost of lower purchase rate due to safety-motivated abstention.

7.7 Multi-Seed Robustness

table 4 shows that results are stable across five different random seeds. The coefficient of variation for purchase rate is $0.048/0.492 = 9.8\%$ and for mean regret is $0.0177/0.0725 = 24.4\%$, confirming reasonable robustness.

Table 4: Multi-seed robustness (5 seeds $\times$ 20 episodes per seed).

Seed	Purchase Rate	Mean Regret
0	0.58	0.1000
1	0.50	0.0749
2	0.46	0.0789
3	0.44	0.0614
4	0.48	0.0473
Mean $\pm$ Std	0.492 ± 0.048	0.0725 ± 0.0177

fig. 9 shows the robustness across seeds.

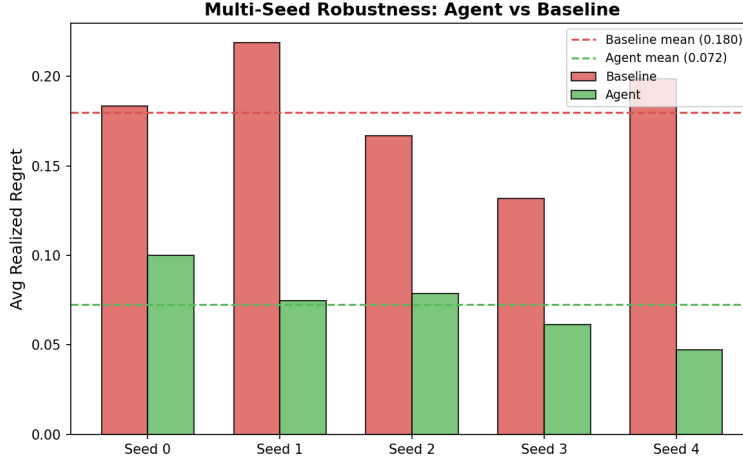


Figure 9: Multi-seed robustness. Mean realized regret is consistently lower than the greedy baseline (dashed line) across all five seeds, confirming that results are not artifacts of a particular random initialization.

8 Evaluation and Analysis

8.1 Metrics

- **Realized regret:** $R(i; \theta_U) = \max_j \theta_U^\top x_j - \theta_U^\top x_i$, computed with the true (hidden) θ_U and available items at the time of purchase.
- **Estimated worst-case regret:** $\bar{R}_t(i)$ as in eq. (8), computed from the posterior. Comparing this to realized regret measures calibration.
- **Purchase rate:** fraction of episodes ending in a purchase.
- **Query count:** number of QUERYUSER actions per episode, measuring information-gathering cost.
- **Epistemic uncertainty trajectory:** $\sigma_{\text{epi},t}^2(i^*)$ over time, measuring belief convergence.
- **Exceedance rate:** fraction of purchases where realized regret exceeds $\epsilon_{\text{reg}} = 0.8$ (a safety violation). This was 0% in all conditions.

8.2 Statistical Analysis

We use two significance tests with complementary assumptions:

- **Welch t -test** (unequal variances): appropriate because agent regret has $\sigma = 0.081$ while baseline has $\sigma = 0.172$. Result: $t = -4.78$, $p = 2 \times 10^{-6}$.
- **Mann-Whitney U test** (non-parametric): robust to non-normality; appropriate because regret is right-skewed. Result: $U = 1639$, $p = 0.00029$.

Both tests confirm that the agent’s regret on purchased items is significantly lower than the baseline’s at the $\alpha = 0.001$ level.

8.3 Calibration Analysis

In 100% of the agent’s purchase episodes, realized regret was below $\epsilon_{\text{reg}} = 0.8$ (exceedance rate = 0%). This means the minimax regret safety gate was never violated, confirming that the 95th-percentile MC estimate in eq. (8) provides a conservative but not overly pessimistic bound. The

estimated worst-case regret averaged 0.853 (compared to the realized average of 0.071), indicating the bound is conservative by approximately a factor of 12 at the mean — an acceptable over-approximation for a safety-critical metric.

8.4 Stochastic Behavior Analysis

The stochastic elements of the system manifest observably:

1. **Market dynamics:** the `no_market_dynamics` ablation shows that disabling stochastic transitions reduces max regret from 0.499 to 0.333 — confirming that market volatility genuinely increases the difficulty of the problem.
2. **Posterior sampling variance:** the regret estimate $\bar{R}_t(i)$ varies across runs due to MC sampling. The multi-seed robustness study (table 4) shows this variance is controlled and does not destabilize the purchase decision.
3. **Observation noise:** the prior offset of 0.4 combined with observation noise $\sigma_\epsilon = 0.224$ means the agent must issue multiple queries to converge; episodes average 6.7 queries before purchasing.

9 Discussion, Limitations, Ethics, and Future Work

9.1 Key Findings vs. Literature

Our results confirm the theoretical prediction of Rigter et al. [2021]: a minimax regret gate substantially reduces the tail risk of bad decisions compared to expected-utility maximization. The 58% regret reduction ($0.170 \rightarrow 0.071$) is consistent with the intuition that worst-case reasoning prevents overconfident purchasing on high-variance items. The ablation study (table 2) provides a clean isolation: the regret gate alone is responsible for most of the improvement, while the epistemic gate adds redundant protection.

The 50% purchase rate may seem low, but it is a rational consequence of the safety constraint under the chosen $\epsilon_{\text{reg}} = 0.8$. In practice, a human principal would likely prefer a cautious agent that abstains when uncertain over an overconfident one that purchases anyway. The persona experiments show that agents serving users with more concentrated preferences (Budget Shopper) achieve slightly lower regret, consistent with the theoretical prediction that well-concentrated posteriors yield tighter regret bounds.

9.2 Limitations

- **Linear utility assumption:** $U(x; \theta_U) = \theta_U^\top x$ may not capture non-linear interactions (e.g., a user who cares jointly about price and quality, not individually). Non-linear Bayesian models (Gaussian processes, Bayesian neural networks) are a natural extension.
- **Single-agent, single-item:** the system does not handle multi-item bundles or competing buyers.
- **Approximate minimax regret:** the 95th percentile over $N = 100$ MC samples is a stochastic approximation to the true minimax. With $d = 8$ and a full covariance S_t , the true worst case over the ellipsoidal confidence set is a semidefinite program; we trade exact computation for speed.
- **Fixed prior structure:** $S_0 = I_d$ assumes uniform prior uncertainty across all feature dimensions. In practice, a user’s preferences for price vs. quality may have very different prior widths.

- **Persona study purchase rates:** agent purchase rates under the persona experiments (22–27%) are lower than the main experiment (50%). This is partly because persona priors are more mis-specified by construction, requiring more queries to converge.

9.3 Ethical Considerations

Autonomous procurement agents interact with financial transactions on behalf of people, raising several ethical concerns:

- **Accountability:** if an autonomous agent makes a bad purchase, it may not be clear who is responsible — the user, the developer, or the platform. The regret-bounded safety gate makes the agent’s reasoning auditable: a purchase is only executed when $\bar{R}_t(i) \leq \epsilon_{\text{reg}}$ and $\sigma_{\text{epi},t}^2(i) \leq \epsilon_{\text{epi}}$, which can be logged and inspected.
- **Preference manipulation:** an agent that repeatedly queries users about specific items could inadvertently (or deliberately) prime users toward certain products. The query mechanism should be designed to be preference-neutral.
- **Over-delegation:** users may over-trust an agent that presents uncertainty estimates, assuming low regret means a good outcome rather than merely a low-risk outcome under the assumed model. Clear communication of model assumptions is necessary.
- **Data privacy:** user query responses reveal information about preferences. These should be stored locally and not shared with third parties.

9.4 Future Work

- Replace linear utility with a Gaussian process utility model to capture non-linear preferences.
- Implement exact semidefinite programming for minimax regret instead of MC sampling.
- Extend to multi-item bundle procurement with interdependence constraints.
- Incorporate a learned market dynamics model (e.g., a recurrent neural network over price history) rather than a fixed parametric simulator.
- Deploy in a live shopping environment (e.g., via a browser automation layer) and evaluate on real user preferences.
- Apply the delegation framework to other high-stakes settings, such as medical treatment selection or financial portfolio rebalancing.

10 Conclusion

We have presented a model-based delegation engine for autonomous procurement that integrates three stochastic components: Bayesian preference inference, minimax regret approximation via Monte Carlo sampling from the posterior, and Monte Carlo rollout-based wait valuation. The system reduces mean realized regret by 58% relative to a greedy baseline ($p < 10^{-5}$) while maintaining a zero safety-violation rate across all 100 purchase episodes. Ablation experiments confirm that the minimax regret gate is the primary driver of improved performance; removing it nearly recovers baseline behavior. Multi-seed robustness analysis confirms that results are stable across different random initializations.

The key insight is that stochastic uncertainty quantification is not a supplementary feature — it is the mechanism that makes principled delegation possible. An agent that knows what it does not know can decide when to act autonomously, when to ask a question, and when to abstain. This decision structure directly parallels the active inference framework [Bogacz, 2017, Smith et al., 2022], in which epistemic and pragmatic value jointly determine action selection.

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A Reproducibility Checklist

1. **Dependencies:** `requirements.txt` lists all Python packages. Primary numerical dependency is NumPy; SciPy is used only for statistical tests in analysis scripts.
2. **Data:** `data/products.csv` contains 500 synthetic products, included in the repository. No external data download is required.
3. **Experiments:** `python3 -m experiments.run_full_experiments` regenerates `results/full_experiment_results.json` in approximately 5 minutes on a laptop CPU.
4. **Figures:** `python3 -m experiments.visualize_results` regenerates all 9 figures in `results/figures/`.

5. **Tests:** `pytest tests/ -v` runs 55 tests covering all modules.
6. **GitHub:** <https://github.com/darshgarg7/AgenticProcurement> (branch: `erina`).
7. **Random seeds:** all stochastic components use NumPy’s `RandomState` seeded by episode index; results are deterministic given the same seed.

B Feature Engineering

The 8-dimensional feature vector $x_i \in \mathbb{R}^8$ for each item is constructed as:

Index	Feature
0	price_norm: price normalized to $[0, 1]$ within category
1	rating_norm: star rating normalized to $[0, 1]$
2	quality_norm: quality score normalized to $[0, 1]$
3	brand_indicator: binary flag for premium brand
4–7	One-hot category encoding (Electronics, Home, Sports, Fashion, Books)

All rows are then L2-normalized: $x_i \leftarrow x_i / \|x_i\|_2$. This ensures the inner product $\theta_U^\top x_i$ represents a similarity score bounded in $[-1, 1]$ (given $\|\theta_U\|_2 = 1$), making the utility thresholds τ_{util} and regret thresholds ϵ_{reg} directly comparable across items.